PAN Validation System Documentation

1. Introduction

This program is a **Python-based data validation system** designed to clean, validate, and analyse PAN (Permanent Account Number) records. It demonstrates how data wrangling, regex validation, and business rules can be applied to ensure data quality. The system processes raw PAN data, removes duplicates, applies validation checks, and generates a summary report in Excel.

2. Program Design

Data Cleaning

- Converts PAN numbers to uppercase and strips whitespace.
- Replaces empty strings with NaN and drops missing values.
- Removes duplicate PAN entries to ensure uniqueness.

Validation Functions

1. **has_adjacent_repetition(pan):**
   Checks if any character is repeated consecutively.
   Example: AABCD → Invalid.
2. **is_sequential(pan):**
   Detects sequential character patterns.
   Example: ABCDE → Invalid.
3. **is_valid_pan(pan):**
   - Ensures PAN length = 10.
   - Matches regex pattern: ^[A-Z]{5}[0-9]{4}[A-Z]$.
   - Rejects adjacent repetition and sequential patterns.
   - Returns **Valid** or **Invalid**.

Output Generation

- Adds a new column Status to mark each PAN as Valid/Invalid.
- Creates a summary table with counts of total, valid, invalid, and missing records.
- Exports results into an Excel file with two sheets:
  - **PAN Validations** (detailed results)
  - **SUMMARY** (aggregated counts)

3. Program Flow

1. Load dataset from Excel.
2. Clean PAN numbers (strip, uppercase, remove blanks/duplicates).
3. Apply validation rules using regex + custom functions.
4. Generate status column for each PAN.
5. Summarize results into a separate DataFrame.
6. Export results to a new Excel file.

4. Example Output

Sample Records:

Pan_Numbers	Status
ABCDE1234F	Invalid
ABCDX1234Z	Valid
AABCD1234E	Invalid

Summary Table:

TOTAL PROCESSED RECORDS	1000
TOTAL VALID COUNT	850
TOTAL INVALID COUNT	120
TOTAL MISSING PANS	30

5. Key Takeaways

- **Regex validation** ensures PAN format compliance.
- **Custom rules** (no repetition, no sequential patterns) enforce stricter business logic.
- **Data cleaning** (removing blanks/duplicates) improves accuracy.
- **Automation with Pandas** makes validation scalable and efficient.
- **Excel export** provides both detailed and summarized results for easy reporting.